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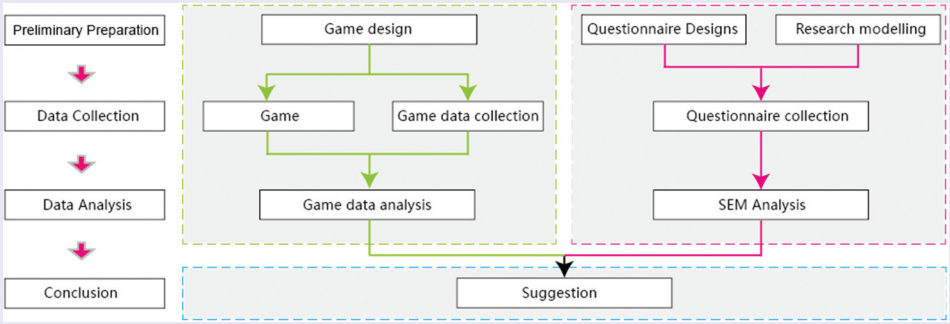
Gamified interaction and urban regeneration: an analysis of serious game participation in rural architectural transformation and its influential factors

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ABSTRACT

This study investigates the utilization of serious games in rural public building design and their impact on residents' contentment with building renovations. It highlights serious games' role as an interactive medium for education, training, and communication, specifically in enhancing residents' approval of rural public building designs. Employing the concept of diegesis from narratology, the study develops a structural equation modelling (SEM) by dividing games into intra-diegetic and extra-diegetic components. This model examines the influence of Game Participation (GP) and Learning Participation (LP) in serious games on satisfaction with updated building designs, as well as the direct and indirect impacts of various factors on these dynamics. The research methodology encompassed creating and applying a serious game focused on architectural design, accompanied by thorough quantitative analysis of the gaming process and results. The study tested multiple hypotheses regarding the positive impacts of game interactivity, perceived creativity, immersive experiences, game quality, atmosphere, and environment on GP and LP. Notably, the beneficial effect of game engagement on learning engagement and the significant positive influence of GP and LP on design satisfaction underscore the potential of serious games in improving educational outcomes and satisfaction with building renovations. Additionally, enjoyment was identified as having a substantial positive moderating effect on the relationships between GP and Design Satisfaction (DS) and between LP and DS, underscoring the importance of enhancing the appeal and engagement of serious games in their design. Additionally, as the game advanced, participants increasingly aligned their opinions on various design aspects, particularly emphasizing the importance of integrating natural landscapes as a key environmental consideration in community redevelopment.



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Serious game; satisfaction; urban regeneration; SEM; architectural design

1. Introduce

The implementation of the policy for volume reduction and development at the government level in 2016 significantly impacted Nanjing's urban landscape. This policy led to a reduction in incremental space, shifting the focus to the redevelopment and utilization of existing space for urban renewal. In the context of this "reduction and dual control" strategy, Nanjing's development predominantly relies on existing construction land, posing substantial challenges for spatial transformation and utilization (Frantzeskaki and Rok 2018). The involvement and satisfaction of residents in the community environment are crucial for the sustainable development of community building renewal (Kamols, Foth, and Guaralda 2021). However, these aspects are often undervalued during the renewal process. Frantzeskaki's 2012 research highlighted that user participation in the design process not only fosters innovation but also promotes sustainable usage (Frantzeskaki, Loorbach, and Meadowcroft 2012). Similarly, Holscher's 2020 study revealed that engaging design stakeholders in the spatial design process can decrease communication costs and lead to co-created

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innovative solutions (Hölscher and Frantzeskaki 2020). Nonetheless, the architectural design industry's inherent barriers may deter stakeholder involvement, as non-professionals face high learning costs and potential resistance to participating in the design process.

Interactive digital games have been recognized as effective tools for enhancing communication, as extensively examined across various disciplines. For instance, Wang et al. demonstrated that games significantly enhance students' attention in classrooms (Wang et al. 2016), Kim et al. investigated their application in business education (2015), and Griffin et al. found that gaming participation improved attention and social skills in adolescents with autism (2021). Within this domain, Serious Games, distinguished by their focus on education, training, and communication rather than mere entertainment (Susi, Johannesson, and Backlund 2007), simulate real-life scenarios. These games facilitate learning or training in virtual environments, tackling complex issues. They offer an innovative approach to engaging residents and stakeholders. By participating in these games, residents gain a clearer understanding of design concepts, actively contribute suggestions, and articulate their needs. This process fosters early consensus and reduces the costs associated with later project revisions (Schalbetter et al. 2023). Awan et al. investigated the efficacy of gamification in junior architectural education by utilizing an online database comprising previous projects and papers on gamification, game design, and architectural education. Specifically, the study examined how game design principles could effectively teach complex interdisciplinary skills. The research highlighted that integrating gamification and game design principles could lead to the development of tools, the gamification of existing curricula, and enhancements in retention, engagement, and mastery of intricate technical skill sets essential for junior architects. This is achieved through the incorporation of elements such as augmented reality, narrative design, and enjoyment (2022). Albano et al. explored employing Digital Pedagogy (DP) for game-based architectural education in China, focusing on enhancing student engagement through Digital Game-Based Learning (DGBL) with HTML5 Package (H5P) (2023). Their qualitative case study assessed the engagement of 121 university students after integrating DGBL into the architecture curriculum. They found a significant increase in student engagement, evident across behavioural, affective, cognitive, and social dimensions. This enhancement included heightened behavioural engagement, increased interest and enjoyment, deeper cognitive engagement, and enhanced interactions with teachers and peers. Building upon previous research, it was observed that while digital games prove effective in architectural design education, the precise mechanism underlying their efficacy remains unclear.

Serious Games transcend mere amalgamations of recreational and educational elements, embodying a complex interplay of game and learning participation. Consequently, an analysis of game participation structures should be contextualized within this participatory process (Susi, Johannesson, and Backlund 2007). Firstly, we designed a set of serious games about architectural design based on the existing site in She Village, Nanjing and invited participants to play the games. In addition to this we built a model to investigate the effects of game participation (GP) and learning participation (LP) in serious games on the satisfaction of rural public building design. The model also examines the direct influence of Game Interactivity (GI), Learning Autonomy (AL), Immersive Experience (IE), Game Quality (GQ), Game Atmosphere (GA), and Perceived Environment (PE) on GP and LP, as well as their indirect effects on design satisfaction. Through a comprehensive study of residents' involvement in Serious Games and its correlation with community building renewal satisfaction, this research aims to offer innovative strategies to enhance the quality and sustainability of community building renovations. Additionally, it seeks to provide valuable insights for urban development and community building.

This paper is structured as follows: First, the design of the game and questionnaire will be developed through internal team discussions and engagement with residents. Second, we will establish the model structure and construct the research model, drawing upon prior studies. Third, residents, architects, design commissioners, and stakeholders will be invited to participate in the game and complete the questionnaires. Fourth, the research hypotheses will be validated and analyzed using Structural Equation Modeling (SEM). Finally, the paper will conclude with a discussion of the research findings (Figure 1).

2. Hypotheses

Isabelle Kniestedt et al. developed a processual engagement theory model for serious games, integrating educational and entertainment aspects (2022). They conceptualized games as a unique category of narrative texts, employing diegesis from narratology. According to their analysis, serious games feature intra-diegetic elements such as immersive environments and interactive creations, while extra-diegetic elements include NPC actors, out-of-game media, and physical environments. Building upon Kniestedt et al.'s theoretical framework, this study introduces nine research variables, divided into intra-diegetic and extra-diegetic categories. The intra-diegetic variables comprise Game Interactivity (GI), Learning Autonomy (AL), and Immersive Experience (IE). The extra-diegetic variables include Game Quality (GQ), Game Atmosphere (GA), and Perceived Environment (PE).

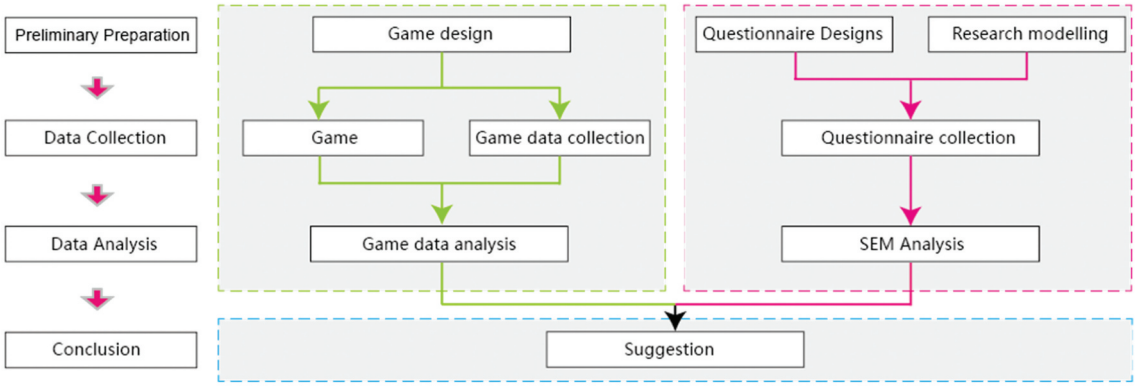


Figure 1. Research framework.

2.1. Inter-diegetic

Game Interactivity (GI) is pivotal in serious games, enhancing both game and learning engagement. Serious games, through the integration of elements such as real-time feedback, diverse interactive tasks, virtual dialogues, and personalized interactions, foster an engaging educational environment. There was a significant correlation between the interactive attributes of the game and the participants' playing and learning experience (Zimmerman 2004). As early as 2005, Moreno et al. showed that educational games that are more interactive are better at keeping participants entertained and learning at the same time (Moreno and Mayer 2005). Ritterfeld et al."s 2009 study indicated that the multimodal nature and immersive interactivity of serious games significantly improve educational quality (Ritterfeld et al. 2009). Chittaro et al., in 2015, found that GI in emergency preparedness serious games notably influences participants" risk perception and self-efficacy (Chittaro and Sioni 2015). Anastasiadis highlighted that GI in educational games effectively communicates the designer's intentions, increasing both game and educational engagement (Anastasiadis, Lampropoulos, and Siakas 2018). Wouter et al. showed that serious games have a significant positive effect on education by analysing the literature related to serious games (2009). Furthermore, personalized interactions, by making learners feel more acknowledged, augment their engagement with game tasks and interest in architectural design knowledge.

Self-motivated players, driven by intrinsic interest and curiosity, tend to be more engaged in the gaming process, rather than being solely motivated by external rewards such as scores or prizes (Loh, Sheng, and Ifenthaler 2015). Guillén-Nieto et al."s study found that the efficacy of video games in teaching and learning is significantly influenced by students" self-driven willingness to learn (2012). This heightened engagement often results in prolonged gaming sessions and more thorough exploration. Learners motivated from within typically delve deeper into game content, enhancing their comprehension of complex concepts or skills (2012). Mohd et al. showed that participants' willingness to self-involvement contributed significantly to the learning component of serious games, as well as serious games contributed significantly to participants' ability to self-learn (Mohd et al. 2019). Such players are likely to be proactive in seeking information, contemplating problems, and applying newfound knowledge, which contributes to enhanced cognitive absorption and memory retention.

The enhancement of perceptual creativity in gaming encourages players to generate innovative design solutions, thereby diversifying game tasks and boosting player interest and participation. Agogué et al. demonstrated that serious games create a communicative, creatively charged environment conducive to knowledge exchange and cooperative learning (Agogué, Levillain, and Hooge 2015). Boyet et al. observed that brainstorming sessions during children's gameplay effectively bolster creative thinking. Additionally, they noted a significant positive impact of creativity in children's serious games on both educational and entertainment aspects (Boyet et al. 2021). Libe et al., in a 2020 study, designed a serious game for children and discovered that while creativity markedly improved the game's entertainment value, it required more time for children to grasp the educational content (2020). Engaging with creative elements in games allows learners to better understand and appreciate the innovation in architectural design. This form of creative interaction fosters a deeper comprehension of design thinking and the creative process, thereby enhancing learning depth.

2.2. Exter-diegetic

Fabricatore et al. in 2007 identified gameplay and game quality as fundamental elements of game design

(Fabricatore 2007). High game quality is essential for player engagement (Liu, Wang, and Lee 2021; Vargas et al. 2014). Liu et al. found that game quality, information feedback, and autonomous motivation significantly influence continued usage of educational game platforms. Additionally, they noted that game quality and autonomous learning motivation indirectly impact learning effectiveness and entertainment engagement (2021). Abdellatif et al. conducted evaluations of several serious games and concluded that enhanced game quality improves both comprehensibility and entertainment value (Abdellatif, McCollum, and McMullan 2018). Elements such as aesthetically pleasing board design, clear rules, and smooth gameplay contribute to the game's attractiveness (Abdellatif, McCollum, and McMullan 2018). Consequently, this paper posits that superior game quality increases players' willingness to engage deeply with the game, thereby facilitating a greater understanding of architectural design principles.

Game atmosphere significantly influences the dynamics of game interaction (Ribeiro et al. 2020). Fotovatnia et al. established that learning motivation positively impacts the atmosphere in classroom games, finding that groups experiencing a more positive educational game atmosphere demonstrate greater willingness to engage in classroom activities and self-directed learning (Fotovatnia and Namjoo 2013). Esposito et al. identified the game atmosphere as a critical, yet often overlooked, component in the design of serious games (2005). Mortara et al. discovered that both the game and social atmospheres of existing games, especially when enhanced by virtual reality technology in cultural heritage contexts, considerably augment participants' learning and entertainment experiences (Mortara et al. 2014). This engaging atmosphere not only heightens the game's appeal but also motivates players to immerse themselves more deeply in architectural design learning tasks. The creation of virtual environments with a realistic feel significantly boosts player engagement in the game scenarios.

The perception of architectural design and environment in games can pique players' interest in game tasks and objectives, thereby increasing their participation. Orland et al. found that the environments in serious games play a crucial role in boosting the learning atmosphere and fostering class cohesion post-gameplay (Orland et al. 2014). Madani et al. observed that game-based learning significantly improves sustained learning ability, with game environments positively influencing this outcome (Madani, Pierce, and Mirchi 2017). Similarly, Dunwell et al. reported that the virtual environments in serious games notably promote continuous learning abilities among participants (2011). Enhanced Perceived Environment (PE) offers a more realistic learning experience. Immersion in these virtual environments enables learners to gain a deeper understanding of the fundamental processes in architectural design, thus facilitating the educational aspect of the gaming experience.

Serious games, which integrate gaming elements and design into non-entertainment domains such as education, training, and health improvement, serve a significant educational purpose (Susi, Johannesson, and Backlund 2007). Alvarez et al. have found that the interactivity and challenge presented in games enhance learners' attention and memory (2011). Furthermore, Mitgutsch et al. reported that immersive gaming environments bolster learners' interest and engagement with the content (2012). This engagement is directly linked to increased satisfaction with design outcomes, leading to higher quality results and more favorable user feedback (Mubin et al. 2022). Consequently, this paper proposes the following hypotheses, illustrated in the hypothesized model depicted in Figure 2:

H1a: GI has a positive effect on GP

H1b: GI has a positive effect on LP

H2a: LA has a positive effect on GP

H2b: LA has a positive effect on LP

H3a: PC has a positive effect on GP

H3b: PC has a positive effect on LP

H4a: PE has a positive effect on GP

H4b: PE has a positive effect on LP

H5a: GQ has a positive effect on GP

H5b: GQ has a positive effect on LP

H6a: GA has a positive effect on GP

H6b: GA has a positive effect on LP

H7a: EM has a positive moderating effect on GP

H7b: EM has a positive regulatory effect on LP

H8: GP has a positive effect on LP

H9: LP has a positive effect on DS

H10: GP has a positive effect on DS

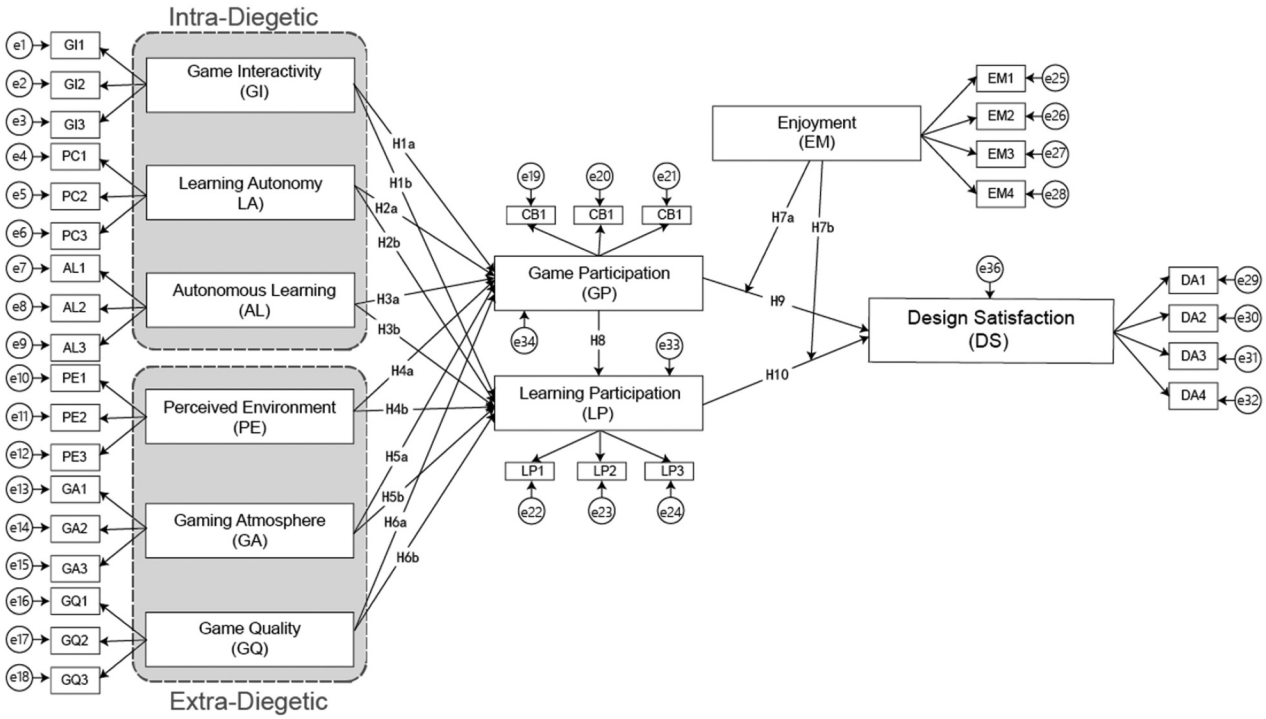


Figure 2. The conceptual framework.

3. Research methodology

3.1. Research design

This game is set in Shecun, located in the Jiangning District of Nanjing, Jiangsu Province, China (Figure 3 (a)). With the national development focus shifting towards rural areas in the last decade, Shecun has experienced an increase in construction investment from both the government and private sectors. Despite this, the control of construction projects remains predominantly in the hands of designers and planners. This has led to a disconnect, with local residents and tourists often not recognizing the value in the new buildings and public facilities. The game specifically targets the most densely populated public building space in Shecun as a case study for its design and research. Its objective is to engage participants in learning about architectural culture and to encourage discussions about Shecun's future development, all while simulating architectural design processes.

3.2. Game design

The game comprises four distinct sections: (1) an introduction to the game rules; (2) collaborative construction of a site board to facilitate a comprehensive understanding of the site's current status; (3) a simulation of architectural design; and (4) the presentation of the final design outcomes, followed by all participants completing a questionnaire.

The game's rules were structured as follows: (1) Game Site and Grouping: The game was set in a public building space in Shecun, Jiangning District, Nanjing, Jiangsu Province, China, specifically chosen for being the area's most densely populated space. This space served as the simulation scenario, as depicted in Figure 3(a) Participants were divided into four groups based on their interests: Designer, Municipal Representative, Local Resident, and Local Cultural Centre, each with distinct roles illustrated in Figure 4(a) (2) Site Board Construction: The game site, designed by the project designer in a grid pattern mirroring the actual site, aimed to create a realistic simulation. Participants collaborated to build the site board, enhancing their understanding of the site's current status and spatial distribution, as shown in Figure 3(b) (3) Architectural Design Simulation: The Designer group's objective was to develop a design-oriented solution that also satisfied the Local Residents. The Municipal Representatives sought to balance user satisfaction with economic and creative goals. The Local Residents considered their satisfaction with the design solutions, aiding the Designers and Commissioners in finalizing these solutions. The Cultural Centre group focused on ensuring cultural appropriateness of the final design. Each game round included unexpected events that impacted players' satisfaction with the architectural design, represented by cards in Figure 4(b) (4) Final Design Results Presentation: During the game's concluding phase, all participants are required to present their final design outcomes, articulating their underlying design

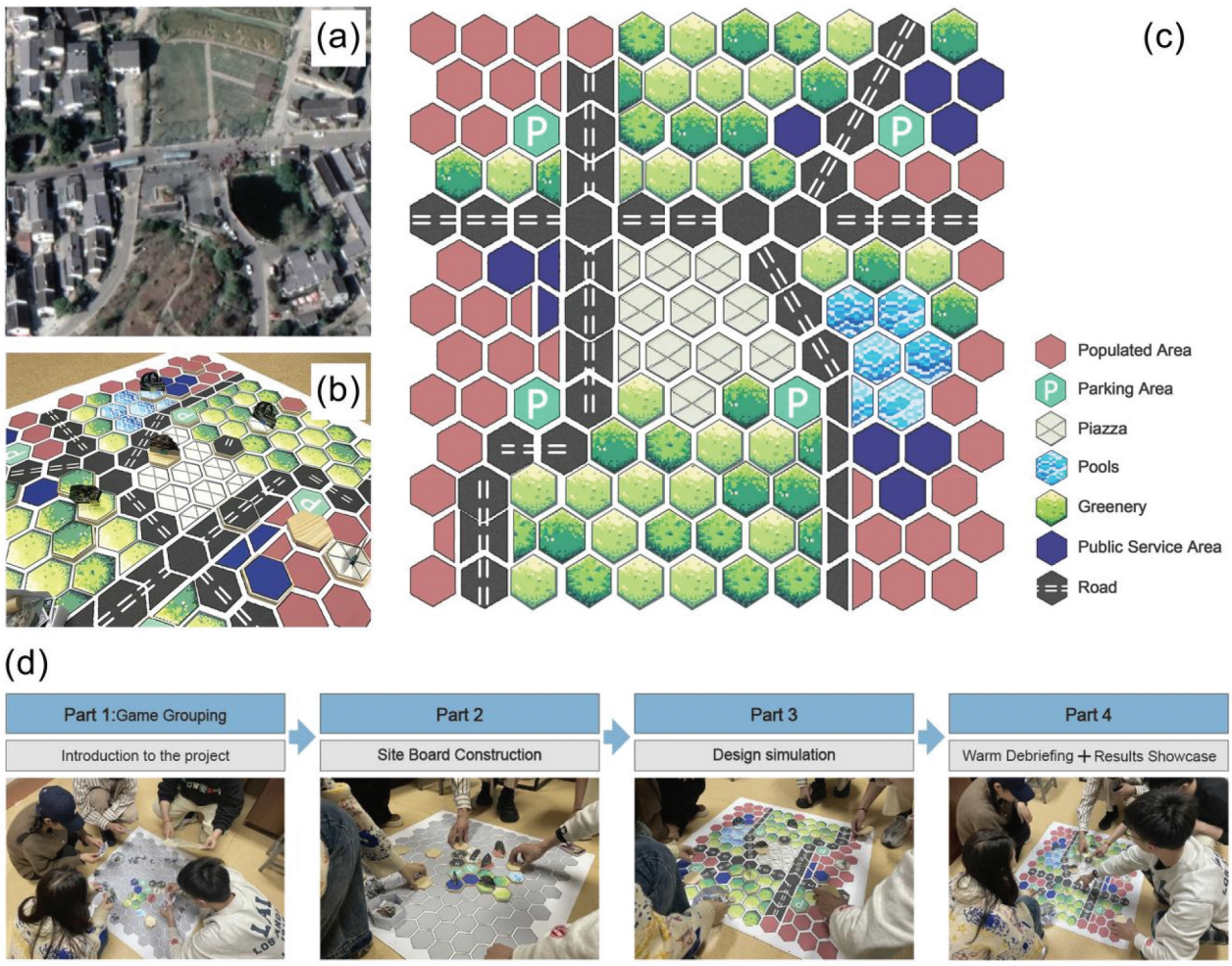


Figure 3. Game tabletop design and game flow chart. (a. Satellite map; b. Stylised translated game desktop; c. Stylised translated game desktop; d. Game flow chart).

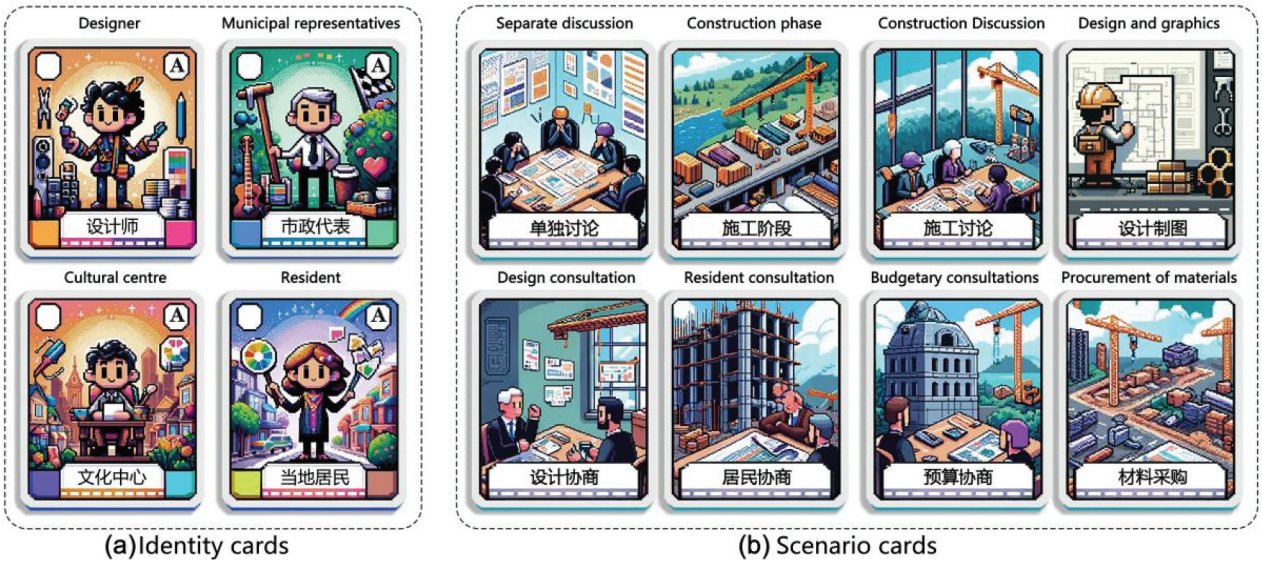


Figure 4. Game card design (a. Identity cards; b. random scenario cards).

Table 1. Descriptive statistics of respondents' characteristics.

Demographic variable	Categories	Frequency	Percentage(%)
Age	16–20 age	120	31.91
	21–25 age	156	41.49
	26–30 age	100	26.60
Gender	Male	216	57.45
	Female	160	42.55
Educational level	Secondary school	60	15.96
	Vocational degree	105	27.93
	Bachelor degree	168	44.68
	Graduate degree	43	11.44
Frequency of board game play	Never played	75	19.95
	At least once a week	54	14.36
	At least once a month	99	26.33
	At least once a half year	148	39.36

philosophy and the rationale behind their decision-making processes. At each game turn, players are required to assess the significance of various building design aspects on a scale from 1 to 10. The criteria for design importance are classified into four categories: design appearance, building utility, cultural significance, and integration with the natural landscape. The game flowchart is shown in Figure 3(d) The design philosophy of the game is centered on using an interactive simulation environment to give participants a deep understanding of the complexities of architectural design and urban planning. By engaging in this manner, the game not only enhances participants' comprehension of the architectural design process but also deepens their engagement with community development. During the gameplay, participants assume various roles such as designers, municipal representatives, and local residents, each with unique responsibilities and objectives. This setup is intended to simulate the negotiation and collaboration among different stakeholders in real-life scenarios. In terms of gameplay, the game is divided into several phases, including establishing a preliminary understanding of the site, collaborative design, simulating the decision-making process, and presenting the final design. This structure helps participants consider issues from multiple perspectives, understanding the needs and challenges of different stakeholders. Additionally, the game incorporates unexpected event cards, adding an element of unpredictability that brings the decision-making process closer to the uncertainties of the real world.

Table 2. KMO and Bartlett's inspection.

KMO		.875
Bartlett's sphericity	spherical test	6500.151
	df-value	496
	p-value	.000

3.3. Sampling strategy

In this study, a total of 386 participants engaged in the game, with 376 completing the questionnaire. As detailed in Table 1, the predominant age group among participants was 21–25 years, constituting 41.49% of the total. Gender-wise, 42.55% of the participants were female, and 57.45% were male. Regarding educational attainment, the majority of respondents, 44.68%, held bachelor's degrees, while only 11.44% were graduate degree holders. When examining experience with architectural design-related tabletop games, 26.33% of the respondents reported playing such games at least monthly. Additionally, 39.36% indicated their participation in tabletop games on at least an annual basis.

4. Analysis and results

4.1. The measurement model assessment

In this study, the Kaiser-Meyer-Olkin (KMO) test yielded a high coefficient of 0.875, demonstrating the data's suitability for factor analysis. The Kaiser-Meyer-Olkin (KMO) test assessed the data's suitability for factor analysis, while Bartlett's test of sphericity yielded a significance value approaching zero, signifying substantial correlation among the data variables. Bartlett's test is a standard procedure for validating data appropriateness in factor analysis by examining correlation matrix calculations. These findings bolster the data's reliability, as depicted in Table 2. Collectively, these results affirm the accuracy of the data analysis and the validity of the findings (Nunnally 1994).

In this study, we employed SPSS version 27.0 to conduct confirmatory factor analysis and descriptive statistics on questionnaire data. Additionally, we utilized structural equation modeling (SEM) via the path analysis software AMOS version 28.0 to analyze research hypotheses, with findings presented in Table 3 (Hair et al. 1998).

Table 3. The values of fit indices.

Fit indices	Chi-square/df	RMSEA	NFI	IFI	CFI
Actual	1.052	0.012	0.945	0.997	0.997
Recommended	<3	<0.06	>0.90	>0.90	>0.90

The Chi-square/df ratio serves as a metric for evaluating the structural equation model (SEM) fit. Specifically, "Chi-square" denotes the model fit chi-square value, while "df" represents the number of parameters in the model. This ratio aids in assessing the reasonableness of model fit. Ideally, the Chi-square/df ratio should range between 1.0 and 3.0, indicating favorable model fit. In our investigation, the ratio stood at 1.052, falling within the recommended range and signifying acceptable model fit. The Comparative Fit Index (CFI) gauges the concordance between observed data and the theoretical model in SEM analysis. A CFI value exceeding 0.90 is conventionally deemed indicative of satisfactory model fit. Here, the CFI attained 0.997, surpassing the 0.90 threshold, thus indicating exceptional fit between the model and the data, as delineated in Table 2. The Normed Fit Index (NFI) functions similarly to the CFI, assessing the degree of alignment between observed data and the theoretical model. Scores range from 0 to 1, with values closer to 1 denoting superior fit. In our study, the NFI value of 0.945 exceeded the 0.90 benchmark, suggesting commendable congruence between the model and the data. The Root Mean Square Error of Approximation (RMSEA) quantifies the residual discrepancy between model predictions and actual data in SEM analysis. Typically, an RMSEA value below 0.08 indicates satisfactory fit. Our computed RMSEA value of 0.012 was markedly lower than the 0.08 threshold, signifying outstanding model fit and minimal residual discrepancy. Moreover, the Incremental Fit Index (IFI) and Tucker-Lewis Index (TLI) are akin to the CFI in assessing model fit, with scores ranging from 0 to 1, where higher values indicate better fit. In this instance, both the IFI and TLI values were 0.997, surpassing the standard criterion of 0.90, further corroborating the model's validity. Collectively, a comprehensive assessment of various fit indices underscores robust model fit, indicative of substantial alignment between the model structure and the data (Hair et al. 1998). This alignment provides a solid basis for subsequent analysis and interpretation.

In this study, data reliability and validity were assessed using Cronbach's alpha, Composite Reliability (CR), and Average Variance Extracted (AVE). Cronbach's alpha measures internal consistency, reflecting the degree of association among a set of items. It ranges from 0 to 1, with higher values indicating greater reliability. In our study, Cronbach's alpha values for all variables ranged from 0.824 to 0.918, surpassing the critical threshold of 0.70 (Nunnally 1994), indicating strong internal consistency. CR, akin to Cronbach's alpha, gauges internal consistency but focuses on construct level rather than individual items. CR values also range from 0 to 1, with higher values indicating greater reliability (Ahmad, Zulkurnain, and Khairushalimi 2016; Eisinga, Te Grotenhuis, and Pelzer 2013; Muilenburg and Berge 2005). Here, CR values ranged from 0.823 to 0.920, indicating robust agreement between variables and affirming their reliability and internal consistency. AVE evaluates the ratio of construct variance to measurement error variance, with values above 0.5 considered acceptable (Fornell and Larcker 1981). Our study recorded AVE values between 0.585 and 0.794, exceeding the 0.5 threshold, suggesting that the constructs elucidate a significant portion of observed variance while maintaining low measurement error. These collective findings underscore the dataset's high reliability and validity. As shown in Table 4.

The tables in this study display standardized regression coefficients, which represent the direct effect values between variables, also known as path coefficients. These tables additionally include details about the statistical significance of these relationships, denoted by the probability values (P-values). A P-value less than 0.001 is marked as '***', a P-value between 0.001 and 0.01 as '**', and a P-value between 0.01 and 0.05 as '*'. For all other instances, P-values are explicitly stated in the respective columns. Employing a significance level of 0.05, the researcher can ascertain the existence and magnitude of a direct effect between two variables, with the standardized regression coefficient quantifying the strength of this relationship. As shown in Table 5.

In our Structural Equation Modeling analyses, game interactivity demonstrated a modest yet significant enhancement in both game engagement (H1a, $p = 0.003$) and learning engagement (H1b, $p = 0.017$), indicating the effectiveness of interactivity in boosting participant involvement. Perceived creativity also had a positive impact on game engagement (H2a, $p = 0.016$) and learning engagement (H2b, $p = 0.006$), suggesting that creative elements within the game foster deeper engagement. The substantial influence of immersive experiences on game engagement (H3a, $p = 0.004$) and learning engagement (H3b, $p = 0.002$) highlights the critical role of immersion. Additionally, perceived environment (PE) demonstrated a negative significant impact on both game engagement (H4a, $p = 0.009$) and learning engagement (H4b, $p = 0.028$), which is a notable deviation from previous expectations. Game quality (GQ), however, showed a significant positive correlation with both game engagement (H6a, $p = 0.006$) and learning engagement (H6b, $p = 0.012$), underscoring the vital role of high-quality game design in participant satisfaction with building renovation. The game atmosphere continued to significantly enhance both game and learning engagement (H5a, $p = 0.008$, H5b, $p = 0.03$), emphasizing the importance of a well-crafted game environment. The study confirms that increased involvement in the game directly improves learning engagement (H8, $p = 0.011$), thereby potentially boosting the educational impact of serious games. The significant positive effects of game engagement (H9, $p < 0.001$) and learning engagement (H10, $p < 0.001$) on satisfaction with building renewal validate the role of educational games in enhancing this

Table 4. Results of confirmatory factor analysis.

Construct	Items and contents	Factor loading	AVE	CR	Alpha
Design Satisfaction	DS1: The functional and aesthetic design of the locally remodelled building met my expectations.	0.840	0.585	0.849	0.846
	DS2: Local building modifications reflect respect and understanding of local culture and history.	0.719			
	DS3: Local building alterations harmonise with their surroundings.	0.762			
	DS4: Satisfied with the quality of local building renovation projects.	0.733			
Game Participation	GP1:I am actively involved in competing and interacting with other players in the game.	0.818	0.608	0.823	0.823
	GP2:I felt entertained by this game during the game.	0.760			
	GP3:I am actively involved in the architectural design and chess challenges in the game.	0.760			
Learning Participation	LP1:By playing this board game, I learnt knowledge and information related to architectural design.	0.793	0.646	0.846	0.846
	LP2:The game improved my skills in architectural aesthetics and strategic planning.	0.818			
	LP3:By playing this game it helps me to learn about local architecture related knowledge.	0.800			
Game Interactivity	GI1:The game's interaction with architectural elements adds to the fun and challenge of the game.	0.827	0.676	0.862	0.846
	GI2:The game provided a wealth of interactive opportunities and allowed me to gain a deeper understanding of architectural design.	0.834			
	GI3:The game's interface makes interacting with game elements smooth and intuitive.	0.805			
Learning Autonomy	LA1:I can independently learn and master the rules and strategies of the architectural board game.	0.954	0.794	0.920	0.918
	LA2:When facing new challenges in the game, I actively seek solutions rather than rely on others' guidance.	0.872			
	LA3:I prefer to explore different architectural design options in the game on my own, rather than imitating other players' strategies.	0.843			
Immersive Experience	IE1:When I play this board game, I feel as if I am actually in a world of architectural design.	0.893	0.704	0.877	0.876
	IE2:The architectural design elements of the game completely immersed me in the experience.	0.819			
	IE3:The game's visual and sound design enhanced my immersion and made me feel more engaged.	0.803			
Game Quality	GQ1:This board game excels in architectural design presentation and game performance.	0.763	0.610	0.824	0.824
	GQ2:The game is rich in content and offers a high-quality architectural design experience.	0.775			
	GQ3:The stability and smoothness of the game allowed me to enjoy the game without interruptions.	0.805			
Gaming Atmosphere	GA1:The architectural design elements of the game create an appealing gaming atmosphere.	0.857	0.683	0.866	0.864
	GA2:The game's social elements and architectural design theme combine to provide an enjoyable interactive experience.	0.840			
	GA3:The game's environment is designed to encourage players to communicate creatively within the context of architectural design.	0.780			
Perceived Environment	PE1:The physical environment I was in provided a comfortable space suitable for playing this game.	0.824	0.710	0.880	0.880
	PE2:I think the physical environment of this game is great.	0.844			
	PE3:I am enjoying the physical environment of this game.	0.860			

Table 5. Standardised regression weights for testing hypotheses (***$p < .001$).

			Estimate	S.E.	C.R.	P	β
GP	<—	GI	0.171	0.057	3.009	0.003	0.183
GP	<—	AL	0.114	0.047	2.406	0.016	0.148
GP	<—	IE	0.141	0.049	2.850	0.004	0.170
GP	<—	PE	−0.140	0.053	−2.616	0.009	−0.156
GP	<—	GA	0.131	0.05	2.638	0.008	0.149
GP	<—	GQ	0.173	0.062	2.775	0.006	0.163
LP	<—	GI	0.138	0.058	2.395	0.017	0.140
LP	<—	AL	0.131	0.048	2.746	0.006	0.160
LP	<—	IE	0.157	0.05	3.135	0.002	0.179
LP	<—	PE	−0.118	0.054	−2.197	0.028	−0.125
LP	<—	GA	0.109	0.05	2.169	0.030	0.117
LP	<—	GQ	0.158	0.063	2.506	0.012	0.141
LP	<—	GP	0.182	0.072	2.539	0.011	0.172
DA	<—	GP	0.268	0.070	3.808	***	0.258
DA	<—	LP	0.399	0.068	5.898	***	0.407

satisfaction. The final results of the study are presented in Table 3 and Figure 5.

4.2. Moderating effect test

This study employed the PROCESS 3.3 plug-in for SPSS and Johnson-Neyman analysis to examine changes in the effect values along the "Game Participation-Design Satisfaction" (GP-DS) and "Learning Participation-Design Satisfaction" (LP-DS) paths as influenced by the moderator variable, Enjoyment. We conducted comprehensive quantitative analyses of these moderating effects to test hypotheses H7a and H7b. The results demonstrated that Enjoyment exerted a significant positive moderating effect on both the GP-DS and LP-DS paths. Specifically, the

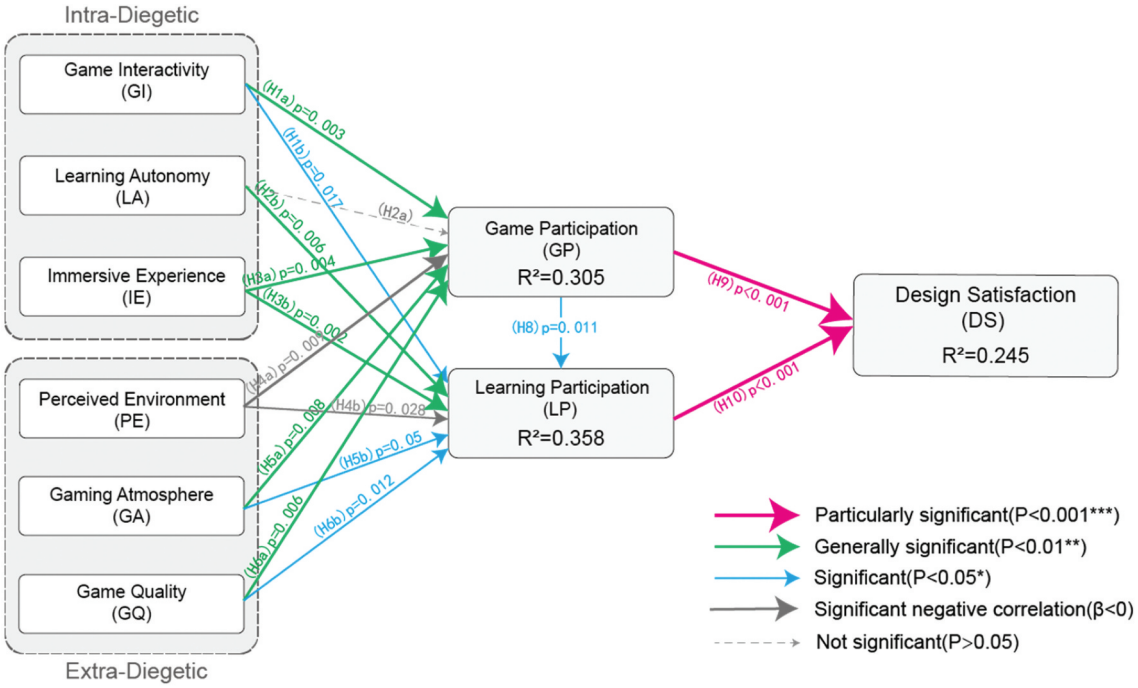


Figure 5. Path analysis of research model.(The dotted lines indicate non-significant paths. *$p < 0.05$; ***$p < 0.001$)

unstandardized coefficients for the interaction terms were 0.1193 for H7a and 0.1663 for H7b, with 95% confidence intervals of 0.099–0.2267 (H7a) and 0.0717–0.2609 (H7b) respectively. Importantly, these confidence intervals do not include zero, thereby providing support for both H7a and H7b. Furthermore, the R-squared value was 0.0099, indicating a strong explanatory power of the overall model.

Figure 6(a) depicts the influence of Learning Participation (LP) on Building Design Satisfaction, categorized into two trajectories based on Enjoyment levels (low and high). Under low enjoyment conditions, an increase in learning engagement corresponds with a rise in building design satisfaction. Notably, this upward trend is more pronounced with high enjoyment, supporting hypothesis H7b and indicating that participant satisfaction with building design escalates more rapidly with greater enjoyment in the learning process. Figure 6(b) demonstrates the correlation between Game Participation (GP) and Building Design Satisfaction, again divided into two trajectories according to enjoyment levels. Consistent with learning engagement, in scenarios of low enjoyment, satisfaction with building design increases alongside game engagement; however, this satisfaction is substantially more significant in the context of high enjoyment, validating hypothesis H8a. A common observation from both charts is that enhanced engagement, whether in gaming or learning, generally elevates satisfaction with architectural design, irrespective of the level of enjoyment. Nevertheless, this increase in satisfaction is markedly more distinct among participants who derive greater enjoyment from playing and learning. These

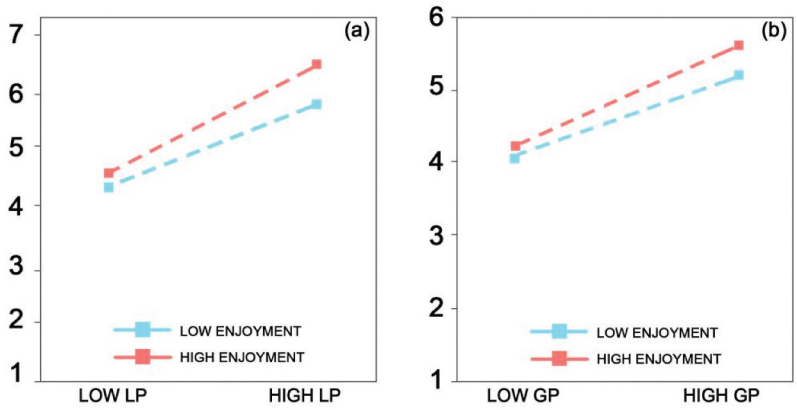


Figure 6. Moderated effects plots (Figure a represents the effect of enjoyment on building design satisfaction when learning participation (LP) is low to high; Figure b represents the effect of enjoyment on building design satisfaction when game participation (GP) is low to high).

findings suggest that in the design of serious games, enhancing game appeal and fostering engagement are crucial for boosting educational effectiveness, particularly for participants who find joy in these activities.

5. Discussion

This study elucidates the efficacy and pivotal factors influencing the use of serious games in rural public building design through meticulous structural equation modeling analysis. Key findings and recommendations are summarized as follows: Game Interactivity (H1a, $p = 0.003$; H1b, $p = 0.017$) significantly enhances both game and learning engagement, suggesting that increasing player interaction can deepen understanding of architectural design processes. Perceived Creativity (H2a, $p = 0.016$; H2b, $p = 0.006$) in the game design boosts players' engagement by fostering creativity through unique architectural challenges and innovative solutions. Immersive Experience (H3a, $p = 0.004$; H3b, $p = 0.002$) in a realistic gaming environment enhances players' interest and engagement in design and learning. Contrary to previous expectations, Perceived Environment (PE) has a negative significant impact on both game engagement (H4a, $p = 0.009$) and learning engagement (H4b, $p = 0.028$), indicating that certain aspects of the game environment may detract from the gaming and learning experience. Game Quality (GQ) shows a significant positive correlation with both game engagement (H6a, $p = 0.006$) and learning engagement (H6b, $p = 0.012$), highlighting the importance of high-quality game materials and well-structured rules. Game Atmosphere (H5a, $p = 0.008$; H5b, $p = 0.03$) positively influences engagement by nurturing creative thinking and exploration in architectural design. Finally, the significant positive effects of game engagement (H9, $p < 0.001$) and learning engagement (H10, $p < 0.001$) on satisfaction with building renewal validate the role of educational games in enhancing this satisfaction. This study offers several recommendations for future serious game designs in architectural design, based on its findings. First, it is evident that game interactivity markedly boosts engagement in both the gaming and learning aspects, thereby enhancing satisfaction with architectural renovation. This implies that incorporating more interactive elements among players or between players and game components could deepen players' comprehension of architectural design processes and techniques. Second, allowing participants greater creative autonomy in the game can spur their creativity, further engaging them in both gameplay and the learning process. Additionally, immersive experiences, as a crucial facet of game design, positively influence participant engagement and the educational outcomes of serious games. Enhancements to game graphics, external factors, and the gaming environment could amplify player immersion, thus elevating their interest and involvement in architectural design and learning. Moreover, the importance of high-quality game design cannot be overstated. Supplying superior game materials and clearly defined rules significantly improves the gaming experience and learning results. Creating a game atmosphere that nurtures creative thinking and exploration in architectural design, for instance, by prompting players to share design ideas, can also boost their engagement in both gameplay and learning. While the perceived environment may have a negative effect on play and learning engagement, this underscores the necessity for designers to thoughtfully craft the game environment to ensure it positively impacts players' experiences and learning processes.

This study explores the positive moderating role of Enjoyment in Serious Game Participation on the relationship between Game Participation (GP) and Learning Participation (LP) with Design Satisfaction (DS). Utilizing SPSS and AMOS software for data processing, along with the PROCESS 3.3 plug-in for SPSS and Johnson-Neyman analysis, the study examined how GP and LP affect Architectural Design Satisfaction (DS), particularly under the influence of Enjoyment. The findings reveal that Enjoyment significantly positively moderates both the GP-DS and LP-DS paths. The interaction terms for these paths have unstandardized coefficients of 0.1193 and 0.1663, respectively, with their 95% confidence intervals not containing zero, thereby validating hypotheses H7a and H7b. The coefficients of determination (R) for the overall path model are 0.234 and 0.0099, signifying the model's robust explanatory power. Graphical analyses, as depicted in Figures 7a and 6b, further corroborate these findings by demonstrating a notable difference in the impact of participation (both GP and LP) on DS at varying levels of Enjoyment. The enhancement of DS is more pronounced under conditions of high Enjoyment, emphasizing the importance of increasing game appeal and engagement in the design and implementation of serious games. This is especially true for participants who experience greater enjoyment. Consequently, game designers should prioritize augmenting the fun and interactivity of games to boost user interest, engagement, educational effectiveness, and satisfaction.

Figure 7 presents the variations in importance ratings assigned to four architectural design aspects in a serious game: design appearance, architectural utility, cultural significance, and natural landscape integration. A comprehensive analysis of the trends for each role is as follows: Designer Role: There is a noticeable upward trend in the emphasis on design appearance as the game progresses. Ratings for architectural utility and natural landscape integration fluctuate slightly but exhibit an overall increasing trend. Cultural significance initially declines but stabilizes later, indicating a growing

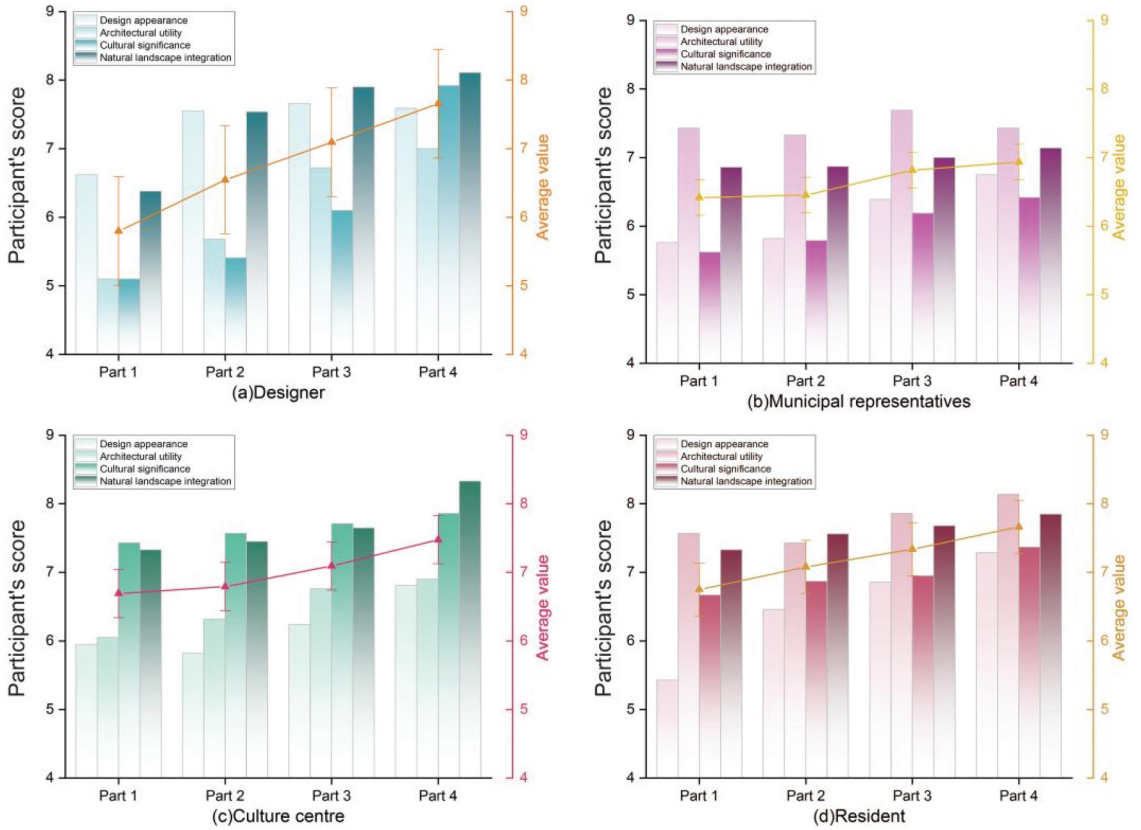


Figure 7. Visualisation chart of the scores given by the participants in this game to the design appearance, architectural utility, cultural genes, and importance of the natural landscape (Fig. a shows the data indicators for the designer player; Fig. b shows the data indicators for the municipal representative player; Fig. c shows the data indicators for the cultural centre player; Fig. d shows the data indicators for the Resident player).

focus on cultural elements in the later stages. Municipal Representatives: This group consistently rated building utility highly, while design appearance received relatively lower ratings, albeit with a gradual increase over time. Cultural significance saw a marked rise in the game's second phase, and natural landscape integration ratings also increased in the latter stages, reflecting a growing importance placed on these elements. Cultural Centres: Ratings from this group show an increasing emphasis on design appearance, particularly in the final stages. Architectural utility received steady ratings throughout, while cultural significance experienced a notable increase in the second phase. The surge in natural landscape integration ratings towards the end suggests a heightened concern for the harmonious combination of nature and architecture. Residents: These participants showed a significant rise in design appearance ratings in the latter half of the game. Architectural utility ratings were consistent, and the emphasis on cultural significance and natural landscape integration grew in the final phase, indicating an increased appreciation of cultural and natural elements in architectural design. Upon thorough analysis of the ratings provided by participants assuming the four roles regarding the significance of four aspects of architectural design, we noted a reduction in the standard deviation of player ratings during the fourth stage compared to the initial three stages. This indicates a tendency for players to converge on similar ratings for the architectural design factors as the game progresses, particularly notable among those assuming the role of Designers. Additionally, we observed a unanimous emphasis from all participants on the importance of integrating natural landscapes into architectural designs. This underscores the substantial value placed on natural environments in the architectural preferences of all players. As a result, forthcoming community redevelopment initiatives in the area should prioritize environmental conservation and the adoption of eco-friendly materials. In conclusion, the game significantly contributed to enhancing stakeholders' comprehension and evaluation of architectural design importance. Through gamified interactions, participants acquired a more comprehensive understanding of design aesthetics, functionality, cultural significance, and natural harmony, fostering a greater appreciation for architectural diversity. This, in turn, facilitated collaboration and dialogue among community members in urban planning, underscoring the practical utility of games in both educational and community engagement contexts.

6. Limitation and outlook

This research encounters several limitations in examining the application of serious games to rural public building design. Primarily, as the focus is largely on the Nanjing region, the findings might not wholly represent all rural areas, thus potentially limiting their generalizability and transferability. Secondly, the specific serious game design employed in this study, tailored for building design simulation, may not be directly applicable to different types of public building projects or to varied cultural and social contexts. Moreover, the study's reliance on quantitative data might have insufficiently captured the participants' personal experiences and subjective perceptions, and the data analysis techniques employed may not fully elucidate complex causal relationships. Lastly, given the study's relatively short duration, it may not have effectively captured the long-term impacts and effects.

Future research in the realm of serious games applied to rural public building design could adopt several strategies to expand and deepen our understanding. Firstly, conducting studies across diverse regional and cultural contexts would not only enhance the generalizability and transferability of the findings but also investigate the effectiveness of various serious game types in public building design and their impact on different stakeholder groups. Secondly, integrating quantitative and qualitative research methods would offer a more holistic view of participants' experiences, emotions, and feedback, and how these aspects influence design satisfaction. This approach represents a critical trajectory for future investigations. Additionally, longitudinal studies to monitor the continued influence and efficacy of serious games would be invaluable. Exploring the incorporation of emerging technologies, such as virtual reality and augmented reality, into serious games could also enrich the gaming experience and learning outcomes, providing more comprehensive insights and practical applications for this field.

7. Conclusion

This research model, grounded in Isabelle Kniestedt et al. s theory of processual game participation in serious games, examines the roles of Game Participation (GP) and Learning Participation (LP) and their impact on satisfaction with the design of rural public buildings. The study also delves into the direct and indirect effects of Intrinsic Storytelling (Inter-Diegetic) and Extrinsic Storytelling (Extra-Diegetic) elements on GP and LP. For this study, the game rules for the architectural design simulation were specifically tailored to Nanjing's local context, and a venue board was developed, providing a valuable resource for future research in this field. The model's fit is demonstrated through robust fit indicators presented in the results section. Additionally, the research identifies a significant positive moderating effect of Enjoyment on the GP-DS and LP-DS paths, as detailed in the section discussing the moderating effects test. These findings offer substantial evidence and insights into the application of serious games in urban planning and design, serving as a reference for further studies in this area. Finally, we counted the ratings provided by players on the importance of the four aspects of building design. We found that the player-game process enhanced participants understanding of the importance of multiple aspects of building design.

Disclosure statement

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